National BULK BAG

A BUYER'S GUIDE TO

UN CERTIFIED BULK BAGS

BY
NATIONAL BULK BAG

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What Makes a UN Certified Bulk Bag Different?

Bulk bags are a versatile bulk packaging option that have the ability to transport and store a variety of products and materials. In certain situations involving the storage and transportation of hazardous materials, a standard FIBC won't meet safety requirements that are necessary for the safe handling of these products/materials.

UN Certified bulk bags are specifically designed to transport and store hazardous materials. In order to be properly classified as UN certified bulk bags they must also meet the specifications and testing standards set out by the UN. These include, but are not limited to:

1. Identifcation & labeling requirements
2. Specifications on max volume
3. High testing standards

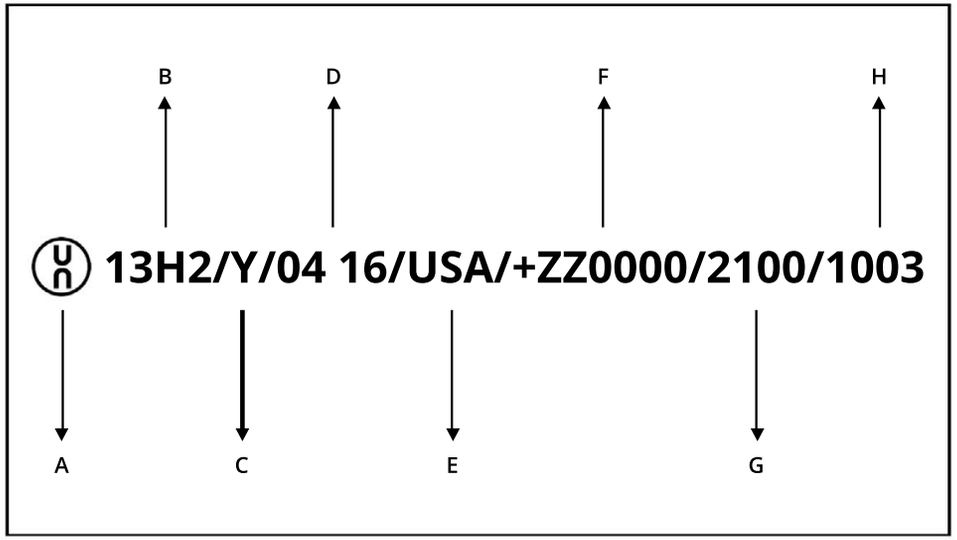
By following the UN specific testing standards, being properly labeled, and properly tagged, a FIBC/bulk bag can be sold as a UN Certified bulk bag.

UN Bulk Bag ID Tag Break Down

UN Bulk Bags have a unique bag labeling system that allows the user to track the origin, materials, manufacturer, coating and other important attributes of the individual UN bag. This labeling system is extremely important to safe handling, filling and use of a UN bag given the hazardous nature of a UN bag's content.

On the middle of this page is an image of what a typical UN bulk bag tag would look like. On the right side of this page is a breakdown of what each part of the tag means and it's significance to UN Certified FIBCs.

The ID tag can appear in 1 or 2 lines of text on the bag and must be at a minimum of 12mm in height.



A. United Nations Packaging Logo - 178.503(e)(1)(ii)

This is the offical UN logo for UN Certified bulk bags. This starts the identification tags.

B. Code number designating FIBC design type - 178.702(a)

This tag can designate features like inner liner, discharge type, etc.

C. Capital letter identifying the performance standard under which the design type has been successfully tested

Different UN Certified bags can be tested under different standards - this designates the bag's testing standards.

D. The numerical month and year of manufacture

The manufacturing date of the FIBC/bulk bag.

E. Country authorizing allocation of the mark - 178.503(e)(1)(ii)

Example: The letters 'USA' indicate the FIBC was manufactured and marked in the United States in compliance with provisions in sub chapter C.

F. The name and address or symbol of the manufacturer

Can also indicate the approval agency certifying compliance with subpart N and subpart O of section 178.

G. The stacking test load in kilograms

The amount of weight the specific bag has been load tested for.

H. Maximum gross mass

Maximum rated combined weight of the materials & FIBC they are packed in.

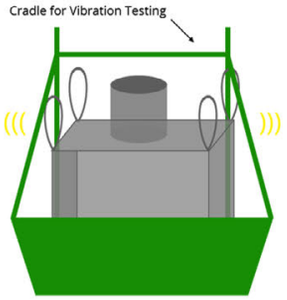
UN Bulk Bag Rigorous Testing Procedures

To ensure that these bags are safe for handling hazardous materials, each bag goes through rigorous testing procedures. The following is a breakdown of each test. We also examine what UN certified bulk bags must go through each test.

For consistency purposes, there are strict standards for testing each bag including the preparation of the bag before it is tested. Proper bulk bag testing preparation includes:

- Bag must be filled no less than 95% full
- Contents must be evenly distributed
- Drop tests must be performed with the solid to be transported OR a non-hazardous material that has the same physical characteristics

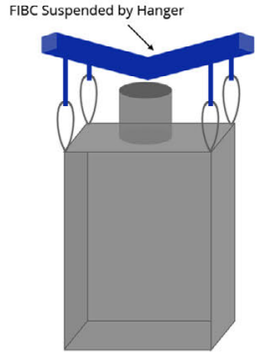
1. Vibration Testing



UN certified bags must be designed to withstand a rigorous vibration test. The test is 60 minutes long and the vibrations are set at a frequency which causes the bag to raise from the vibration table platform.

All UN Certified bags must be capable of passing this test. The criteria for passing this test is no rupture of the bag and no leakage of contents.

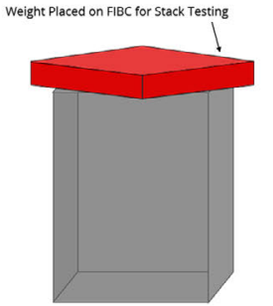
2. Top Lift



During the top lift test, the bag is lifted by an approved lifting device. During the test, the bag must maintain its integrity with a 6:1 SWL load for 5 minutes which means the bag must hold 6-times the safe working load the bag is rated for the duration of the test. The criteria for passing for this test is no permanent deformation or loss of contents.

The top lift test is required for all FIBCs that are designed to be lifted from the top and/or the side.

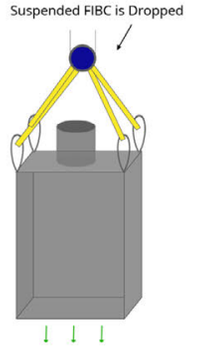
3. Stack Testing



During the stack test, a top load is placed on a filled bag for 24 hours. The top load weighs 1.8-times the combined maximum permissible gross mass of the bags that may be stacked on the FIBC. The passing criteria for this test is no loss of contents and no deterioration of the bag making it unsafe for transportation.

The stacking test is required for all FIBCs that are designed to be stacked.

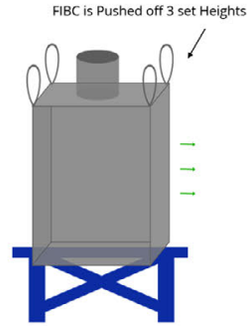
4. Drop Testing



During the drop test, the bags are filled in accordance with proper UN testing preparation. The bag is then dropped from specific heights based on the packaging group (PG I: 70.9", PG II: 47.3", PG III: 31.5") on the most vulnerable part of the base of the bag. The criteria for passing the drop test is no loss of contents and only a slight momentary discharge is permitted.

All UN Certified Bulk Bags must pass the drop test.

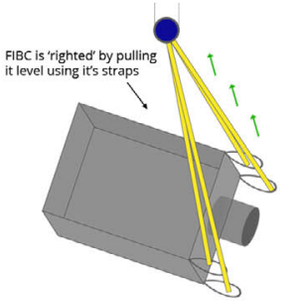
5. Topple Testing



During the topple test the bag is toppled from a specific height based on the packaging group (PG I: 70.9", PG II: 47.3", PG III: 31.5") and will need to be toppled onto any part of its top on a smooth surface. The criteria for passing the topple test is no loss of contents and only a slight momentary discharge is permitted.

All UN Certified bags must pass the topple test.

6. Righting Testing



During this test, the bag is on its side and lifted to an upright position. The bag is then lifted at a specific speeds and must be able to be lifted by both one (bags with two lifting straps) or two (bags with 4 lifting straps) approved lifting devices. Criteria for passing this test is no damage to the FIBC or lifting devices making them unsafe for transportation or handling.

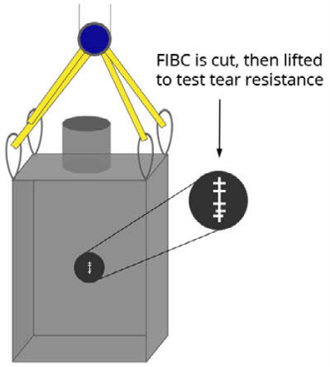
The Righting Test is required for all UN Certified Bulk Bags that are designed to be lifted from both the top and/or the side.

7. Tear Test

During the tear test, the bag wall is punctured with a knife at a 45-degree angle halfway between the bottom of the bag and the top level of the contents. The tear test is split into two parts. Part one of this test is a uniform top load with twice the maximum permissible safe working load. To pass part one, the bag must remain intact for no less than five minutes with a 45-degree knife cut.

For part two of this test, the bag must maintain its integrity with a 45-degree knife cut when being lifted and remain intact suspended for no less than five minutes. The criteria for passing this test is the knife cut cannot grow more than 25% of its original length.

The tear test is required for all UN Certified bulk bags.



Thanks for Reading!

We hope you found this UN Certified Bulk Bag Buyer's Guide helpful and informative. It is important to us at National Bulk Bag to provide valuable and educational materials on the various products we proudly sell.

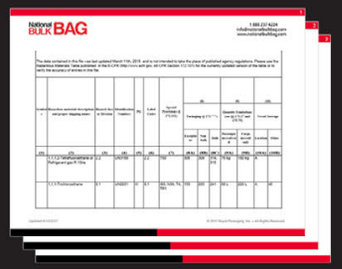
Through discovery and our detailed custom bulk bag solution process, our UN experts work with you to find the right bag for your hazardous product/material. We know that every customer and application is unique and want to help you determine if one of our in-stock UN bulk bags or custom manufactured bulk bags is best for your specific needs.

Schedule Time to Talk

Our certified UN bulk bag experts are ready to provide you with:

- UN Bulk Bag knowledge and expertise.
- An in-depth analysis of the FIBC that you are currently using.
- An in-stock bulk bag solution (if applicable)
- A custom manufactured bulk bag solution at one of our premier manufacturing facilities.
- A warehousing solution that suits your unique needs.
- Price quote for your custom FIBC solution.

SCHEDULE TIME TO TALK



Download the 'Hazardous Materials Chart'

Get access to over 600 pages of information on hazerdous materials. This includes important information regarding labeling codes,packaging information, and any special provisions that apply.

This is a great resource when considering if you need a UN Certified FIBC.

CLICK TO DOWNLOAD THE CHART